

Depth phenomenology of truncated Weil quadratic forms: fifteen L -functions, one ladder

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Abstract

For the semi-local Weil quadratic form on the window $[-\log \sqrt{\mu}, \log \sqrt{\mu}]$ (Connes–Consani framework), we measure the smallest eigenvalue $\lambda_{\min}(\mu)$ for ζ and fourteen of the seventeen real primitive Dirichlet characters of conductor $q \leq 24$, with matched, depth-adequate Galerkin bases up to $\mu = 38$. Four structural facts emerge. (i) The depth obeys an *affine law* $-\ln \lambda_{\min} = s\mu + b$ for the entire family, ζ included ($s_\zeta = 11.7 \pm 0.2$; the previously apparent bend was a basis artifact); linearity in μ is the signature of the integer lattice — any continuum mechanism caps at $O(\log \mu)$. (ii) The drilling speeds s , spanning 0.46 to 11.7, are described at $\sim 6\%$ by an empirical four-variable map in $(\gamma_1, \gamma_2 - \gamma_1, D, \text{parity})$; the map survived two adversarial preregistered tests and failed a third (χ_{17} , small first gap $\gamma_2 - \gamma_1$), so it is presented as a map, not a law. Counting-type functionals are ruled out wholesale: the zero-counting deficit of the desert is quasi-universal (≈ 0.5 missing zero for every character), so *depth is a bandwidth phenomenon in absolute frequencies, not a counting one*. (iii) The quasi-null ladders of all fifteen L -functions share one architecture: rung spacings collapse onto a single spacing-vs-level profile $\Delta(\ell)$ ($\pm 12\%$, ζ indistinguishable from the characters at equal level); a dilated-Slepian reading of this profile was proposed and then dissolved by its own precision campaign. Radical spectroscopy reveals the microscopic organization: *rung k recruits the $(k+1)$ -th χ -supported prime*, in order, with tower signs modulated by $\chi(p)$. (iv) Positivity itself is a *quorum* of the Euler product: every proper sub-product violates it by $O(1)$ while the complete product lands on the razor’s edge (ζ at $\mu = 11$: 3.58×10^{-48} in our basis) — certified computationally at fixed windows ($\mu = 11, 16, 22$, and χ_3 at $\mu = 11$) in the companion paper, which also reduces the all-scales statement to two harmonic-analysis assertions. All claims here are measurements with stated error bars; every code path, zero cache and artifact is public.

1 Setting

For $\mu = e^L$ and a real primitive character $\chi \bmod q$ (or ζ , $q = 1$, with its pole term), the semi-local Weil form Q_χ acts on even test functions supported in $[-L/2, L/2]$: archimedean term in a closed Frullani representation on $[0, L]$ (so the finite- T archimedean artifact of the literature is structurally absent), minus the χ -signed prime sum over $p^k \leq \mu$, plus, for ζ , the pole term. The family was assembled incrementally during the campaign; the three odd characters of discriminant $-8, -20, -23$ were not computed and are the natural next out-of-sample tests (§6). Matrices are assembled in the shifted-cosine basis with the exact correlation table (derived and certified in the companion paper [6], App. A–B: the form is *exactly* Weil’s pairing, $Q(f) = \sum_\gamma |\hat{f}(\gamma)|^2$, no normalization factor), and diagonalized in multiprecision (45–110 digits). Every matrix is validated against the zero side of the explicit formula (40–280 zeros per L -function, computed independently).

Two systematic effects are controlled throughout, both documented in the lab notebook [7]: *transients* (narrow-desert characters approach their asymptotic slope slowly, from below, so slopes at small cutoffs are lower bounds — corrections up to $+18\%$), and *basis depth demand* (an under-sized basis inflates $\lambda_{\min}$ by orders of magnitude and fakes a downward bend of deep ladders; ζ ’s apparent non-linearity was exactly this

artifact, and its ladder is linear, $s_\zeta = 11.7 \pm 0.2$, intercept ≈ -20 — reconciling the Connes–Consani $\sim 10\mu$ heuristic, which is this affine law read at $\mu = 11$).

2 The map

χ	q	parity	γ_1	$\gamma_2 - \gamma_1$	D	s
χ_{24}^o	24	odd	1.977	2.746	3.174	0.46 ± 0.02
χ_{24}^e	24	even	2.689	2.604	3.174	0.50 ± 0.03
χ_{19}	19	odd	1.516	3.961	0.876	0.58 ± 0.03
χ_{21}	21	even	2.315	3.465	2.683	0.58 ± 0.03
χ_{17}	17	even	3.728	1.907	0.907	0.71 ± 0.05
$\chi_{15}^\dagger$	15	odd	3.057	2.286	2.802	≥ 0.82 (rising)
χ_{13}	13	even	3.119	4.112	0.984	0.95 ± 0.04
χ_{12}	12	even	3.805	2.888	3.174	1.01 ± 0.02
χ_{11}	11	odd	2.477	4.323	1.035	1.07 ± 0.04
χ_8	8	even	4.900	2.728	1.673	1.53 ± 0.02
χ_7	7	odd	4.476	2.370	1.183	1.70 ± 0.05
χ_5	5	even	6.648	3.183	1.302	2.47 ± 0.02
χ_4	4	odd	6.021	4.223	1.673	3.04 ± 0.02
χ_3	3	odd	8.040	3.209	1.501	4.00 ± 0.10
ζ	1	(pole)	14.135	6.887	0	11.7 ± 0.2

Table 1: Asymptotic slopes of $-\ln \lambda_{\min} = s\mu + b$, uniformized at $\mu = 30$ –38: s is the secant slope over the two segments ending at $\mu \approx 30$ and $\mu \approx 38$, whose agreement ($\pm 2\%$ typical) is the stationarity criterion. $D = \sum_{p|q} \log p / (\sqrt{p} - 1)$ is the prime mass removed from the window by the conductor. Intercepts b are not yet uniformized across the family ($b_\zeta \approx -20$) and are omitted. $\dagger \chi_{15}$ is still transient at $\mu = 38$: its lower bound is excluded from the asymptotic slopes and from the fit, and its settling scale is not yet bracketed.

The empirical description. *Protocol*: the training set is the twelve characters of Table 1 minus the two final preregistered tests (χ_{21} , χ_{17}); χ_{19} was itself a preregistered test of an earlier round, then absorbed; χ_{15} enters with its transient-corrected estimate; ζ is outside the map (pole; anomaly ratio $\rho \approx 1.3$ against the even baseline). On these twelve points,

$$\ln s = 1.32 \ln \gamma_1 + 0.45 \ln(\gamma_2 - \gamma_1) - 0.13 D + 0.28 [\text{odd}] - 1.95,$$

RMS 6.1%, leave-one-out median 4.8%. Its stress record, all preregistered: γ_1 alone and “prime conductors stay high” were *killed* by χ_{19} ($\gamma_1 = 1.516$, the narrowest desert, predicted 0.28 resp. ≈ 0.9 , measured 0.58); the twin pair mod 24 (equal removed primes, opposite parities) established the arithmetic-density variable against γ_1 (decisive pair χ_{11} vs χ_{24}^e : nearly equal γ_1 , slopes 1.07 vs 0.50) — and the anti-transient campaign designed to test this contrast *widened* it; χ_{21} landed at +9% of its preregistered prediction; χ_{17} missed by −26%, exactly where the gap variable extrapolates below its fitted range ($\gamma_2 - \gamma_1 = 1.91$), and the refit on fourteen points moves the exponents ($0.45 \rightarrow 0.68$) — parameter instability. Honest status: *a map accurate to $\sim 10\%$ with charted failure modes, not a closed law*.

3 Bandwidth, not counting

Every counting-type candidate for the “single variable” died, and the autopsy is the finding. The desert normalized by local zero spacing fails the decisive pair; the Laplace count $L(\tau) = \sum e^{-\gamma_k/\tau}$ conflates density with landscape (ζ predicted at 39.6 against 11.7); and the *vacancy* $V(\tau) = \int (\bar{N}_\chi - N_\chi) e^{-t/\tau} dt$ — built from the exact Γ -phase smooth counting, whose calibration reproduced $c \approx 0$ for all fourteen characters and $c = 1.006$ for ζ , the classical +1 — turns out to be *quasi-universal*: $V \in [-0.10, +0.05]$ across a family whose s spans a factor 25. In counting units every desert holds the same $\approx \frac{1}{2}$ missing zero, because the first zero sits where the smooth count reaches $\frac{1}{2}$; there is nothing to read. The only non-trivial vacancy in

the family is ζ 's ($V = +0.401$: the pole's push, visible at last in a counting observable, yet useless to the model). Meanwhile the variables that work ($\gamma_1, \gamma_2 - \gamma_1$) are *absolute frequencies*: the ladder measures the room, in hertz, where a test function can hide its spectrum — a Slepian time-bandwidth phenomenon, not a statistics of zeros.

4 One ladder, drilled at fifteen speeds

Linearity is the lattice. Since the depth grows like $s\mu$, the effective concentration budget grows *linearly* in μ ; but any mechanism carried by the continuum window — prolates against a fixed band — caps at $O(\gamma L) = O(\log \mu)$, and so does the Landau count of the desert over the multiplicative lattice. Linear depth requires each *integer* of the window to contribute a bounded increment: the drilling is a discrete-lattice phenomenon (the Connes–Consani map E is its embodiment), invisible to continuum analysis alone. A zeroth-order lattice estimate, $c_{\text{eff}} \approx \gamma_1 N_\chi(\mu)/2\pi$, holds to a factor 1.9 ± 0.9 across the factor-25 family (an order-of-magnitude estimate, not a fit) — right scale, wrong details; band and lattice do not factorize (the recurring χ_{17} anomaly marks the coupling).

The universal rung profile. Across 33 rungs of the fifteen quasi-null ladders, the spacing of a rung (in $-\ln \lambda$ units) depends only on its depth level ℓ , not on the character: by level bands, 10.9 ± 1.5 on $[0, 15)$, 13.5 ± 1.5 on $[15, 40)$, 16.0 ± 1.6 on $[40, 80)$, 16.0 ± 2.1 on $[80, 150)$ — ζ 's rungs interleave indistinguishably with χ_3 's and χ_4 's at equal level; fitted profile $\Delta(\ell) \approx 9.8 + 0.65\sqrt{\ell}$ (RMS 1.8). A “dilated Slepian” reading, $\lambda_k \approx (1 - \lambda_k^{\text{Slepian}})^\kappa$ with $\kappa = 2.85 \pm 0.26$, was proposed and then *dissolved by its own precision campaign*: ten-rung ladders fit the Fuchs form beautifully per ladder, but the fitted exponent does not transfer (3.5–5.4 with c free) — the model has no well-defined κ , and “ e or 3?” was ill-posed. What survives, model-free, is the $\Delta(\ell)$ collapse itself: the problem factorizes into a universal architecture (a harmonic-analysis target, blind to arithmetic) times one number per L -function, the drilling speed $s(\chi)$.

The recruitment law. Radical spectroscopy — projecting each component form (pole, archimedean, one tower per prime) onto the bottom eigenstates — reveals the organization: *rung k recruits the $(k+1)$ -th χ -supported prime, in order*, the next prime whispering at the 10^{-2} level one rung before its turn. For ζ the ground state is a pole/archimedean interferometer ($+1.502/-1.443$) trimmed by the 2-tower alone (-0.058), all other primes dead to three decimals; for χ_3 the same staircase runs $2, 5, 7, 11, \dots$ (3 skipped, $\chi_3(3) = 0$) with tower signs modulated by $\chi(p)$ — where $\chi_3(2) = -1$, the 2-tower *supports* the ground state ($+0.345$ against an archimedean -0.347): ζ 's roles reversed. Since each new prime power $p^k \leq \mu$ reinforces the rungs that recruited p , and the Λ -weighted count of prime powers is $\psi(\mu) \approx \mu$, the linearity of the drilling is the prime number theorem seen from the radical.

5 Positivity is a quorum

Recruitment describes the *organization* of the completed form's ladder; its *existence* is a different and sharper fact. Every partial Euler product inside the window has a negative smallest eigenvalue of order 1 — at $\mu = 11$: $\lambda_{\min}$ from -0.52 to -1.15 over all $2^4 - 1$ proper subsets of $\{2, 3, 5, 7\}$, the residual negatives being the radical-side face of the off-line pseudo-zeros of truncated Euler products — and the moment the last interior prime joins, the form lands on the complete 48-digit-deep ladder: a razor's-edge landing at every scale, with depth *not decomposable* into per-prime contributions. This is now a certified theorem at fixed windows ([6]: 340 proper sub-products violated, exhaustively at $\mu = 11, 16, 22$ and for χ_3 without pole, ball-arithmetic witnesses frozen in the repository; the all-scales quorum remains a conjecture), whose mechanism — a prime cannot be diagonally silent on a rung while coupling to it from another without creating a negative direction — reduces the general quorum to two harmonic-analysis assertions (diagonal silence before recruitment; survival of the coupling), neither mentioning RH. One consequence deserves emphasis: since the complete form sits at $e^{-s\mu}$ of zero, *no inequality with slack can prove Weil positivity at these scales*; only an identity — exhibiting the truncated form as the sum of squares it secretly is —

can. Measured, this is the sharpest explanation we know of why generic analytic approaches to RH fail: the margin they would need does not exist.

6 Open problems

Derive the universal profile $\Delta(\ell)$ (the surviving lemma candidate; pure harmonic analysis). Derive $s(\chi)$ (the coupling of window arithmetic to the drill; zeroth order $\gamma_1 N_\chi/2\pi$, structure open). Prove the quorum at every scale via the silence/coupling reduction. And, for the seismograph program: with a $\sim 10\%$ map, the anomaly statistic $\rho(\chi) = s_{\text{meas}}/s_{\text{map}}$ is calibrated (ζ , the one member with a pole, registers $\rho \approx 1.3$ against the even baseline); scanning real even characters at growing conductor is mechanical, the cost growing with the kernel width $\frac{1}{2} \log(3q/\pi)$. (A Siegel zero is a *real* zero, not a small γ_1 : the statistic ρ , calibrated on the critical-line desert, would register an exceptional modulus through the map’s failure, not as a narrow desert per se.) The three uncomputed odd characters ($d = -8, -20, -23$) are the cheapest next tests; χ_{-23} , odd with small γ_1 at nearly the cost of χ_{19} , is worth more than any refinement of exponents.

Reproducibility

Code, zero caches, certified artifacts, and the full lab notebook (seventy-eight journal steps, eight artifact families documented, fifteen preregistered hypothesis executions); repository state as of commit **9785235** (a tagged release will freeze the exact state):

<https://github.com/darreal44/riemann-weil-phenomenology>

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